

CONTROL FLOW STATEMENTS PART-2

ii) while loop with else block:

Syntax:

```
while condition:  
    Statement1  
    Statement2  
  
else:  
    Statement3
```

In the above syntax else block is executed whenever the condition becomes false.

Example:

```
1) i=1  
   while i<=10:  
       print(i)  
       i+=1  
   else:  
       print(i)  
2) i=5  
   while i>=1:  
       print(i)  
       i-=1  
   else:  
       print(i)
```

iii) for loop:

Syntax:

for variable in IterableObject:

Statement1

Statement2

Examples:

1) for i in range(10):

print(i)

2) for i in range(1, 10):

print(i)

3) for i in range(1, 10, 2):

print(i)

4) for i in range(10, 1, -1):

print(i)

iv) for loop with else block:

Syntax:

for variable in IterableObject:

Statement1

Statement2

else:

Statement3

Examples:

1) for i in range(10):

print(i)

```
else:
    print(i)
2) for i in range(1, 10):
    print(i)
else:
    print(i)
3) for i in range(1, 10, 2):
    print(i)
else:
    print(i)
4) for i in range(10, 1, -1):
    print(i)
else:
    print(i)
```

v) Nested loops:

A loop with in another loop is called as nested loop.

The inner loop is executed for each iteration of outer loop.

Examples:

```
1) i=1
   while i<=3:
       j=1
       while j<=10:
           print(j)
           j+=1
       i+=1
```

```
2) for i in range(1, 4):  
    for j in range(1, 11):  
        print(j)
```

Jump Statements:

- 1) break Statement
- 2) continue Statement
- 3) pass Statement

i) break Statement:

It terminates the nearest enclosing loop.

Examples:

```
1) i=1  
   while i<=10:  
       if i==5:  
           break  
       print(i)  
       i+=1  
2) for i in range(1, 11):  
    if i==5:  
        break  
    print(i)
```

ii) continue Statement:

It passes the control to the next iteration of a loop.

Examples:

```
1) i=1  
   while i<=10:  
       i+=1  
       if i==5:  
           continue
```

```
        continue
    print(i)
2) for i in range(1, 11):
    if i==5:
        continue
    print(i)
```

iii) pass Statement:

It passes the control to the next statement.

Examples:

```
1) i=1
   while i<=10:
       pass
       i+=1
   print(i)
2) for i in range(1, 11):
    pass
    print(i)
```

By

Mr. Venatesh Mansani

Naresh i Technologies